Pranay Kumar Pandey

Kharghar, Navi-Mumbai, Maharashtra, India

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Education

Indian Institute of Space Science & Technology, Thiruvananthapuram **2025 – 2027**
M.Tech in Manufacturing Technology, Department of Aerospace Engineering — CGPA : 7.13 *Keralam, India*

Amity University Mumbai **2021 – 2025**
B.Tech in Aerospace Engineering — CGPA: 8.28 *Mumbai, Maharashtra*

St. Andrews Jr. College of Science, Panvel **2019 – 2021**
Higher Secondary Certificate (HSC), 89.7% *Mumbai, Maharashtra*

Convent of Jesus & Mary High School, Kharghar **2008 – 2019**
Secondary School Certificate (SSC), 85% *Mumbai, Maharashtra*

Interest

- Additive Manufacturing
- Wire Arc AM
- Finite Element Analysis
- Propulsion Systems
- Thermal Protection Systems
- Composite Manufacturing
- Research & Development
- Subtractive Manufacturing
- Generative AI

M.Tech Thesis

Experimental and Numerical Investigations on Arc DED **Aug 2026 – Present**
Department of Aerospace Engineering, IIST *Thiruvananthapuram, Keralam*

- Conducting a parametric study of WAAM process parameters in austenitic stainless steel
- Integrating ultrasonic vibration into MIG-based WAAM as a hybrid process to refine grain structure and improve mechanical properties of high-temperature components
- Developing FEM and CFD models at macro scale
- Applying Phase-Field Modelling (PFM) to simulate solidification & dendritic growth at the micro scale
- Validating models through Optical Microscopy, SEM, EBSD, and EDS across different layers

Experience

Vikram Sarabhai Space Centre (VSSC), ISRO **Jun 2026 – Jul 2026**
Intern, Glass & Electronic Materials Division, Advanced Materials Group *Thiruvananthapuram, Keralam*

- Developed high-emissivity borosilicate glass coatings for reusable silica-tile TPS
- Synthesized glass frits, prepared coated specimens and fired them onto porous silica tile substrates
- Characterized thermal behaviour of the frits and stained coatings
- Developed in-house computational toolbox to bridge development timelines in glass formulation
- Preview of toolbox can be at : https://aero-toolbox.vercel.app

FOSSEE, IIT Bombay **Mar 2025 – Jul 2025**
Research Assistant *Mumbai, Maharashtra*

- Mentoring a cohort of interns and fellows on simulation projects, ensuring quality and technical depth
- Collaborating on national workshops and capacity-building programs to scale CFD outreach
- Expertise in Computational Fluid Dynamics (CFD) and OpenFOAM implementation
- Upgrading Case Studies and Migrating to latest version of OpenFOAM

IIT Kharagpur **Jun 2024 – Jul 2024**
Aeroengine Development Intern *Kharagpur, West Bengal*

- Conducted comprehensive parametric analysis of Variable Cycle Engine performance using GasTurb 14
- Performed detailed thermodynamic cycle analysis and evaluated engine performance under various operating conditions
- Evaluated engine performance across take-off, subsonic, and supersonic flight regimes
- Analyzed VABI configurations and their impact on thrust and fuel consumption
- Optimized design parameters including bypass ratios, pressure ratios, and turbine entry temperature

IIT Bombay **May 2023 – Aug 2023**
FOSSEE Summer Fellowship *Mumbai, Maharashtra*

- Conducted numerical study of Non-Newtonian Fluid Flow across Sinusoidal Microchannels using OpenFOAM
- Analyzed velocity profiles, pressure gradients, and viscosity for complex biological and industrial fluids
- Implemented rheological models in open-source and commercial CFD software platforms
- Developed expertise in CFD pre-processing, post-processing, and scientific writing methodologies
- Gained extensive knowledge in literature survey, scientific writing, and pre & post-processing techniques

Skills

- **Programming & Software:** Abaqus, Ansys Fluent, Ansys Workbench, Autocad, C/C++, CNC-Train, Fusion 360, GasTurb, Github, ICEM, Mastercam, MSC Apex, Matlab, OpenFOAM, OpenVSP, Paraview, PATO, Python, Scilab, Simufact Additive, Simufact Welding, Simulink, SolidWorks, Ultimaker Cura
- **Soft Skills:** Leadership, Effective Communication, Public Speaking
- **Languages:** Hindi, Marathi, English, German (A2)

Achievements and Awards

- Emerging Talent in Open Source Software - National FOSSEE Awards 2025
- Qualified GATE AEROSPACE - 2025, AIR 651
- Global Qualifier, NASA Space Hackathon (Oct 2024)
- First Prize, CFD Hackathon by FOSSEE, IIT Bombay (Dec 2023)
- NPTEL Course Topper, Aeroengine Gas Turbine Cycle, IIT Kharagpur (Oct 2023)
- FOSSEE Summer Fellowship, IIT Bombay (May 2023)

Conferences

- International Conference on Advances in Materials and Manufacturing Processes for Strategic Sectors (ICAMPS 2026), September 2026 – The Indian Institute of Metals (IIM) Trivandrum Chapter
- International Conference on Advances in Aerospace & Energy Systems (IAES), April 2024 (ISRO LPSC - IIST, Kerala)
- System Dynamics Theory & Practice, February 2024 (IEOR, IIT Bombay)

Projects Undertaken

- **Development of Random Positioning Machine for Simulating Microgravity on Earth (M.Tech Course Project, 2025):** Working on Model Development and Digital Twin
- **Advancement in Affordable and Accessible Avionics Control Systems for Modern Rocketry (B.Tech Project, 2025):** Developed cost-effective, robust avionics for model rocketry, focused on lightweight structures and reliability.
- **Autonomous Drone Navigation in GPS-Denied Martian Terrain (IROC-U, 2024):** Systems Engineer managing Sensor Fusion, Script Development and adherence to Competition Outcomes.
- **Conceptual Supersonic UAV Design for Military Applications (NACDEC VIII, 2024):** Team Mentor: Contributions to aerodynamic and propulsion system design for supersonic flight.